Solve the radical equation, and check all proposed solutions

1. $\sqrt{x+5} = 7$

Solve and check the equation.

2. $x^{3/2} = 8$

3. $(x+6)^{5/2} - 5 = 3$

Solve the absolute value equation or indicate that the equation has no solution.

4. $|x+6| = 7$

5. $3\,|x+2| + 4 = 2$

Solve the polynomial equation by factoring and then using the zero product principle.

6. $x^3 + 8x^2 - x - 8 = 0$

Determine if the given function is even, odd, or neither.

7. $x^4 + 2x^2$.

Give the domain and range of the relation, and determine if it is a function. [2 pts each]

8. $\{(6,7),(10,-8),(9,6),(1,2)\}$

a. Domain:

b. Range:

c. Function?

9. $\{(5,-4),(1,-3),(1,0),(10,-3),(26,-3)\}$

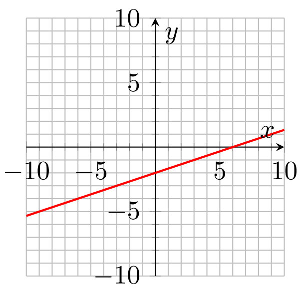
a. Domain:

b. Range:

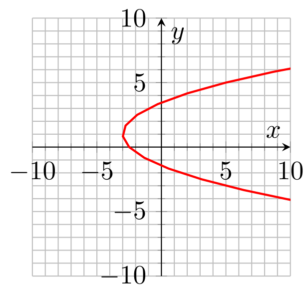
c. Function?

Use the vertical line test to determine whether or not the graph is a graph in which y is a function of x. [2 points each]

10. **a.**

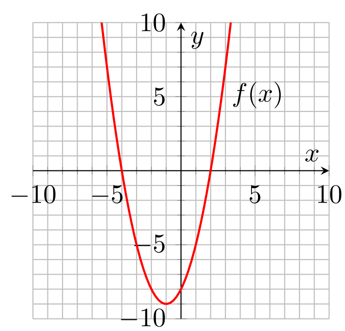


b.



Use the provided graph to complete the problem.

11. [3 points each]



a. Determine the $x-$intercept(s) of $f(x)$.

b. Determine the $y-$intercept(s) of $f(x)$.

c. Determine the domain of $f(x)$.

d. Determine the range of $f(x)$.

Evaluate the function at the given value of the independent variable and simplify.

12. $f(x) = -5x + 1$; $f(-3)$

13. $f(x) = x^2 - x$; $f(x - 2)$

Find the slope of the line that goes through the given points.

14. $(2, -3)$ and $(-7, 8)$.

Use the given conditions to write an equation for the line in the indicated form.

15. [8 points] Passing through $(5, 8)$ with a slope of 3.

a. Point-slope form.

b. Slope intercept form.

16. [8 points] Passing through $(3, 8)$ and perpendicular to the line $y = -\frac{1}{3}x + 7$.

c. Point-slope form.

d. Slope intercept form.